

25-11-14

EP 1750651

7702-EP-EPT

Amended
Description

EP App N° 04 804 019.0

Composition for improving skin, hair and coat health.

The present invention pertains to a composition for preventing, decreasing and/or treating skin and hair/coat disorders or damages, such as is effected by inflammatory reactions, environmental factors, ageing or cancer. In particular, the present invention relates to the use of flavanones compounds or their derivatives in nutritional, cosmetic or pharmaceutical compositions for improvement of human or pet animal skin and coat conditions.

Background of the invention

The most prominent epithelial tissue in living beings is the skin, which represents the largest organ in the organism. The system of skin integument, which comprises the epidermis, dermis and the stratum corneum, correlates with those of internal organs and concurrently interacts with the surroundings. Being the interface between the environment and organism itself, the skin is heavily influenced by external factors and also variable parameters of the organism's inner system. The skin's regulative mechanisms need, therefore, always be active to induce systemic changes necessary to maintain normal pathological events concerning skin integument morphology and activities.

A great deal of processes assuring the adequate consumption of increased affluence of energetic and plastic substances according to the skin's needs become guarantors of morphological and functional stability of skin structures. So, the state of integuments determines the realization of metabolic processes necessary for skin cell viability and activity leading to the presence of healthy skin peculiarities such as barrier function, elasticity, turgor properties, humidity, pigmentation etc..

During the lifetime of a living being different signs, characteristic of ageing, appear on the skin or hair, with the principal clinical signs being the appearance of fine lines and deep wrinkles which increase or are accentuated with age, loss of hair, reduced hair density, glossiness, color, oilness, fiber diameter, etc....

These signs of ageing are even promoted by exposure of the skin and hair to exogenous influences, such as e.g. UV-radiation, pollutants, free radicals or chemical substances.

In the art several means have been proposed to prevent destructive effects of environment or ageing on skin epithelial cells. For example, means to prevent skin deterioration or ageing is to provide compounds scavenging free radicals. In this respect EP 0 761 214 discloses singlet oxygen quenchers comprising aniline derivatives and difurfuryl amine derivatives, which are reported to reduce the oxidative stress to the skin. In this respect, JP 2002 029978 A discloses that the combined use of citrus extracts based on eriocitrin and vitamin C as free radical scavenger.

As another background document, document XP003030654 discloses the use of citrus extract against inflammation without identifying any constituent(s) which might be responsible of this effect.

Although there is a great diversity of active compounds for ameliorating skin and hair or coat health, there still exists a need in the art to provide new active compounds. In particular, an object of the present invention is to provide compositions that may be used over a long period of time by humans or pets, and susceptible to be provided in the form of a nutritional supplement, for example a nutritional composition.

Summary of the invention

In a first aspect, the present invention relates in claim 1 to a nutritional, cosmetic or pharmaceutical composition for oral administration for human or pets, which contains as active compound, at least one flavanone compound or its derivatives, or a mixture thereof, in an efficient amount to prevent, treat or alleviate skin, hair and/or coat disorders and ameliorate skin, hair and coat health selected from hesperetin and hesperetin-7-O-glucuronide for use in reducing cell death.

The present invention also relates in claim 2 to a composition for oral administration, which contains as active ingredient, at least one flavanone compound that is hesperidin for

use in preventing and/or treating inflammation.

In another aspect, the invention provides the use the compounds of claims 1 and 2 ~~of at least one flavanone or its derivatives or mixtures thereof~~, as active compound in the preparation of a nutritional, cosmetic or pharmaceutical composition intended for ~~preventing or treating skin, hair and/or coat disorders, thus ameliorating skin health conditions in humans or pets~~ the specific claimed effects.

The composition according to the present invention may be in the form of a complete nutritional formula, a dietary supplement to be orally administered to a human or an animal, or a compound for pharmaceutical use.

Administering to a human or pet animal, a food composition as described above, results in an improved skin health, e.g. on photoprotection, hydration, dryness, firmness, thickness, elasticity, oilness, regular pigmentation, immunity or hair and coat health, e.g. improving hair and coat gloss, hair density, color, oilness, ameliorating hair fibre diameter, sebum production, glossiness and preventing hair and coat loss. Also, the composition according to the present invention is administered to a human or an animal, for ameliorating antioxidant status, barrier function, to prevent or modulate oxidative status, sebum production or composition, or to reduce signs of ageing. It also helps to reduce risks of cancer or inflammation.

Detailed description of the invention

According to the first object, the invention provides a ~~nutritional, cosmetic or pharmaceutical composition for oral administration for human or pets, which contains as active compound, at least one flavanone compound or its derivatives, or a mixture thereof, in an efficient amount to prevent, treat or alleviate skin, hair and/or coat disorders or damages and thus ameliorate skin, hair and coat health~~ according to claim 1.

A composition according to claim 1 is also provided.

The flavanone compounds of interest are natural glycosides that can be found principally

in fruits from the genus Citrus, such as orange, lemon, bitter orange, grapefruit, for example or in a lesser extend in other vegetables. They are present in majority in the peel of the fruit, but also in large amounts in the pulp and thus also in citrus fruit juice. The compounds according to the present invention may be isosacuranetin, naringin, hesperidin, or eriodictyol, poncirin, neohesperidin, for example, and their its derivatives selected from their aglycone forms, chalcone forms, glycosylated forms or methylated forms namely hesperetin and hesperetin-7-O-glucuronide. Also, their sulfated or The glucuronidated forms which are found as product of metabolism in blood are used.

~~In a last aspect, derivatives may be obtained by several processes known in the art, such as enzymatic treatments.~~ For example, glucose-7-hesperetin is prepared by rhamnosidase or hesperidinase treatments which are known enzymatic treatments.